

neither $f(x) = 2x + 1$
 $f(1) = 3$ $f(-1) = -1$

MAT1375, Classwork4, Fall2026

Ch4. Basic Functions and Transformations

1. Even Functions and Odd Functions.

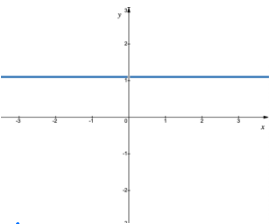
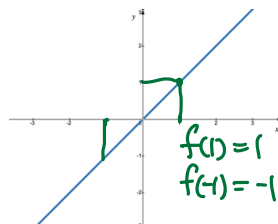
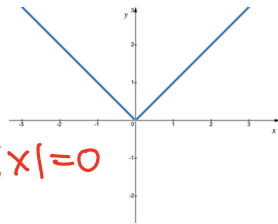
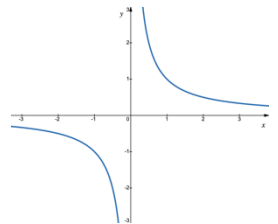
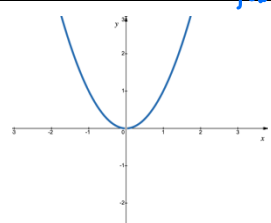
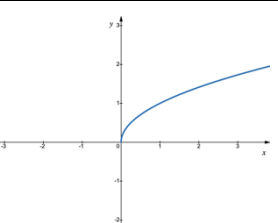
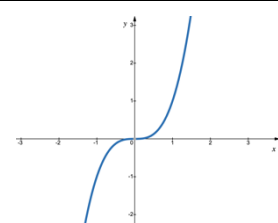
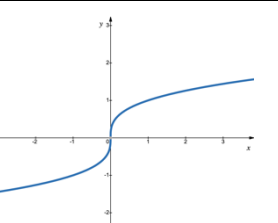
Even function: A function f is called even if $f(-a) = f(a)$ for all a .
 Odd function: A function f is called odd if $f(-a) = -f(a)$ for all a .

2. Complete the table of basic functions:

$f(x) = 1$ $f(x) = c$

$f(x) = x^2$ $f(1) = 1$ $f(-1) = 1$ even

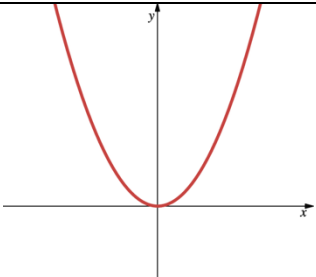
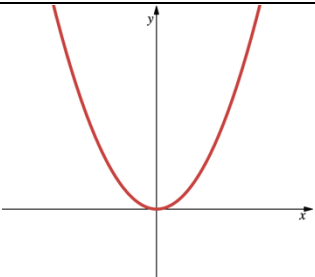
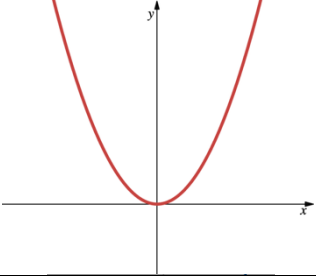
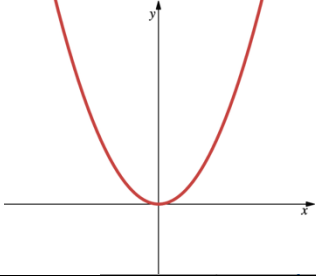
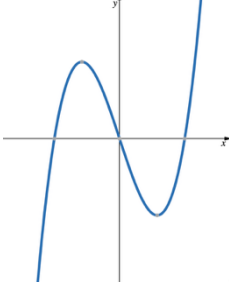
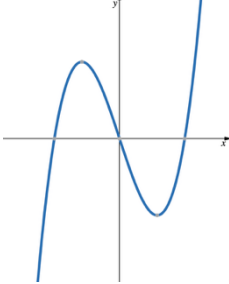
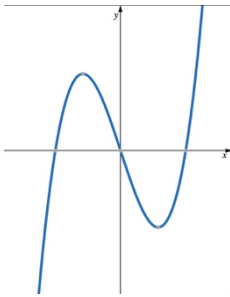
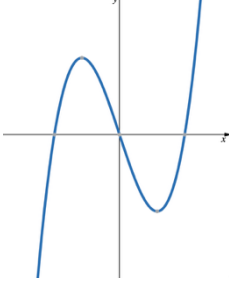
$g(x) = x^3$ $g(1) = 1$ $g(-1) = -1$

$f(x) = \text{constant function}$  All real numbers, $\mathbb{R}$ Domain: $(-\infty, \infty)$ Range: $\{c\}$ or $\{c\}$ Odd or Even: Even $f(-a) = f(a)$	$f(x) = x$ linear function  All real numbers, $\mathbb{R}$ Domain: $(-\infty, \infty)$ Range: $(-\infty, \infty)$ Odd or Even: Odd $f(-a) = -f(a)$	$f(x) = x $ absolute function  $ x = 0$ Domain: $(-\infty, \infty)$ Range: $[0, \infty)$ Odd or Even: Even $f(-a) = f(a)$	$f(x) = \frac{1}{x}$  Domain: Range: Odd or Even
$f(x) = x^2$ parabola function  Domain: $(-\infty, \infty)$ Range: $[0, \infty)$ Odd or Even: Even	$f(x) = \sqrt{x}$  Domain: Range: Odd or Even	$f(x) = x^3$  Domain: Range: Odd or Even	$f(x) = \ln(x)$  Domain: Range: Odd or Even

3. Determine which of the following functions are even, odd, or neither.

(a) $f(x) = 2x^2 + x^4$ (b) $f(x) = x + 3x^3$ (c) $f(x) = x^3 + 4x^2$

4. Complete the table

Transformation of $f(x)$	Given $f(x)$ and then draw the transformation	
_____ $y = f(x) + c$	$c > 0$: 	$c < 0$: 
_____ $y = f(x + c)$	$c > 0$: 	$c < 0$: 
_____ $y = cf(x)$	$c > 1$: 	$0 < c < 1$: 
_____ $y = f(cx)$	$c > 1$: 	$0 < c < 1$: 

_____ $y = -f(x)$	_____ $y = f(-x)$
